

HIMANSHU SWAMI

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PROFILE SUMMARY

Department Rank 1 Computer Science Engineering student at Gandhinagar University (CGPA: 9.75/10), currently engineering Quantum Key Distribution Network systems at ISRO – Space Applications Centre. Proficient in full-stack web development (MERN stack) and Python-based backend systems, with hands-on expertise in post-quantum cryptography, secure distributed architectures, and REST API design. Seeking a software engineering role to apply production-grade backend and security engineering skills.

ACHIEVEMENTS

- Ranked 1st in Department out of the entire CSE batch at Gandhinagar University with a CGPA of 9.75/10.
- Competitively selected for an internship at ISRO – Space Applications Centre (SAC), Ahmedabad, one of India's premier space research institutions.
- Earned HackerRank SQL (Advanced) certification, placing in the top percentile of SQL practitioners globally.

EDUCATION

Gandhinagar University

Bachelor of Technology in Computer Science Engineering

– CGPA: 9.75 / 10.0 | Department Rank 1

Gandhinagar, Gujarat, India

Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: Java, Python, C

Frontend: HTML, CSS, JavaScript, React.js, EJS

Backend: Node.js, Express.js, Flask, FastAPI, REST APIs

Databases: MongoDB, MySQL, Oracle Database

Frameworks & Libraries: NumPy, Pandas, PyQt

Networking & Security: Cryptography, Quantum Networks, QKDN, Secure Communication Systems

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Developer Tools: Git, GitHub, GitHub Copilot, Postman, VS Code, MongoDB Compass

EXPERIENCE

Space Applications Centre (SAC), ISRO

Project Intern – Quantum Key Distribution Network (QKDN)

Ahmedabad, Gujarat, India

Jun 2026 – Present

- Architected and implemented software components for a 4-node QKDN system, enabling quantum-safe key exchange across a distributed, secure communication architecture.
- Integrated 3 post-quantum cryptographic algorithms (Kyber512, SPHINCS+, AES-GCM) into a unified key management pipeline, strengthening resistance to classical cryptanalytic attacks.
- Built FastAPI-based REST APIs for Key Manager nodes across a 4-node distributed system, implementing Argon2id authentication, role-based authorization, and UART/serial communication layers with async-to-blocking bridge patterns.
- Delivered 50+ pages of architectural documentation — covering A* routing, system topology, and login/logout flows — that accelerated team onboarding and design review cycles.

PROJECTS

StayVista | *Node.js, Express.js, MongoDB, Mongoose, Passport.js, Cloudinary, MapTiler, EJS*

- Engineered a full-stack accommodation booking platform with 5+ REST API endpoints, enabling property listing, browsing, and booking with real-time geolocation map integration via MapTiler.
- Secured user authentication with Passport.js role-based access control, reducing unauthorized access risk while supporting concurrent multi-user sessions.
- Optimized media delivery via Cloudinary integration and designed a MongoDB database schema with MVC architecture and session management, improving load performance and data scalability.

Weather Dashboard | *React.js, Express.js, OpenWeatherMap API, Chart.js, HTML, CSS, JavaScript*

- Delivered a real-time weather monitoring dashboard covering 1,000+ cities, surfacing 5+ live metrics (AQI, UV Index, temperature, humidity, wind) via OpenWeatherMap API integration.
- Improved data readability by visualizing weather trends with Chart.js and React hooks (useState, useEffect), yielding a fully responsive UI across desktop and mobile browsers.

CERTIFICATIONS

SQL (Advanced) Certification | *HackerRank*

Python 101 for Data Science | *IBM Developer Skills Network*

Databases for Developers: Next Level | *Oracle Dev Gym*